DataBaseAgent: a local desktop workbench for building and publishing topic-specific biological data repositories with human-gated AI agents

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6 November 2026

Abstract

Agent-curated repositories such as scBaseCount show that large language model (LLM) agents can discover, annotate and reprocess public sequencing data at a scale manual curation cannot reach. That demonstration is a single bespoke repository: its agent, pipeline and cloud deployment were engineered for one assay by one institute. Most researchers need something smaller and more specific, and the need takes one of two shapes: a laboratory that already holds a curated table and wants it published as a searchable resource, or a laboratory that must first gather every public dataset on its organism, tissue or perturbation of interest and process it uniformly with tools it trusts. GPSAdb, a manually curated compendium of 3,048 human gene-perturbation RNA-seq datasets, is the authors' own instance of both. Here we present DataBaseAgent (DBA), a general-purpose desktop workbench that turns the agent-curated repository pattern into a tool any researcher can run, in either shape. DBA ships as one binary for Windows, macOS and Linux with no server or database to deploy for the workbench itself. Website construction is driven by the user's goal, the supplied data and a domain skill: the agent inspects the available results, reads a SKILL.md containing domain knowledge and interpretation constraints, and authors the page structure, visualisations and interactions. The current implementation produces a standalone static HTML site with embedded data; fixed table and card templates remain optional. Preview, iteration and download are local, while publication through a configured Model Context Protocol (MCP) provider is a separate action. The broader design includes dynamic websites, but database-backed services for large datasets remain an implementation and evaluation limitation. A collecting project states its topic, organism, assay and inclusion criteria in free text; candidates are gathered from Europe PMC, PubMed or GEO, or imported; an LLM reviewer proposes include, exclude or uncertain with a verbatim evidence snippet that the application verifies against the record, and a dataset enters the repository only through a recorded decision that carries a written reason and names who made it; the user writes the processing pipeline as ordinary shell steps and locks a version; jobs run on the user's own Linux server over SSH under CPU and concurrency quotas, network failures are retried a bounded number of times and everything else is escalated to the user with the log and a resume point; outputs are checksummed, manifested and archived locally. One project agent and four specialists, each with its own model, instructions and fixed tool list, work in a single conversation per project; they act only through the same validated commands as the interface, run only when addressed, and every call, including every MCP call, is recorded. Any OpenAI-compatible, Anthropic or Gemini endpoint, including locally served models, can be the model. We present DBA as a design contribution: its two workflows and

architecture, the skill and MCP mechanisms and their boundaries, the safeguards around acting agents, two collection walkthroughs on live services, one of which carries a single GPSAdb-style dataset from a GEO search through a help request to a checksummed count matrix, an Agent-authored PBMC5k website case with 5,025 cells, and a feature comparison with repository-scale curation agents, dataset fetchers, screening tools, workflow platforms and static publishing tools. Usability and the quality of the repositories users build with it remain to be evaluated. Software availability is described in the Data and code availability section.

Keywords: data curation, AI agents, human-in-the-loop, reproducible pipelines, static web publishing, agent skills, Model Context Protocol, desktop software, biological databases

1 Introduction

Public sequencing archives hold far more data than any community uses. The Sequence Read Archive (SRA) and the Gene Expression Omnibus (GEO) grow faster than curated atlases can absorb them, and most of their content is reachable only through free-text metadata that was written for submission, not for reuse [1,2]. Uniformly reprocessed resources such as ReCount and recount3, ARCHS4 and refine.bio showed that re-running one pipeline over thousands of public experiments removes processing artefacts and makes cross-study analysis routine [3,4,5]. Each of them, however, is a centrally built product with a fixed scope, and the labour of selecting, annotating and reprocessing datasets is paid once by one team.

scBaseCount moved that labour to an AI agent [6]. Its curation agent, SRAgent, is a hierarchy of ReAct-style language-model agents [7] that queries NCBI tools, decides whether an SRA experiment is 10x Genomics single-cell data, extracts organism, tissue and disease, and stores the result in a relational database; a Nextflow pipeline, scRecounter, then reprocesses every accepted experiment [8]. The outcome is the largest public single-cell repository, more than 502 million cells from 27 organisms, and the authors show that agentic extraction recovers metadata that deterministic tools such as ffq and pysradb cannot [6,9,10]. They also state the general lesson: the approach "provides a blueprint for how AI can be leveraged to autonomously curate biological data repositories" [6]. The blueprint itself, though, is not something a typical laboratory can pick up. SRAgent runs on Google Cloud Run against a hosted PostgreSQL database, scRecounter runs on GCP Batch, the pipeline is specific to 10x chemistry, and the published run consumed an estimated 13.2 trillion tokens at a cost of about 17,000 USD [6].

Most repository needs sit in the long tail below that scale, and the authors' own database is the motivating case. GPSAdb collects human cell-line RNA-seq datasets in which one gene was knocked out or down: 3,048 datasets for 1,458 genes in its published version, each found by searching GEO and ENCODE with knockdown and knockout terms, screened by hand, given its control and perturbed samples by hand, and reprocessed with one pipeline into differential expression tables and 6,096 gene sets that its GPSA tool uses to infer candidate causal perturbations from a user's own expression data [11]. The paper promised an annual survey of new data; the database registry lists its last update as August 2022 [12], because every update repeats the whole manual cycle. Other groups face the same cycle: a plant group wants every public single-cell dataset of a root tissue, a disease group every bulk cohort that matches a phenotype, a perturbation group every screen that targeted a pathway. Today such a group stitches together a search in PubMed or Europe PMC, a spreadsheet of candidates, a fetching tool such as pysradb, ffq or nf-core/fetchngs [13,9,10], a workflow manager [8,14,15], and a cluster account, and then keeps the whole arrangement alive by hand. Platforms such as Galaxy and Seqera cover execution well but start after the datasets have been chosen [16]; screening tools such as ASReview and Rayyan triage literature but stop before any data is touched [17,18]; and general-purpose biomedical agents such as Biomni and CellAgent analyse data already in hand without

addressing which datasets to gather [19,20]. Building the repository is hard. Maintaining it is harder: new datasets keep appearing, the pipeline changes, remote jobs fail at night, and the reasons a dataset was included are lost in a chat log or a colleague's memory.

A second need sits beside the first and is simpler. Many laboratories already hold the curated table that makes a database: GPSAdb's list of datasets with their genes and cell lines, a consortium's annotation of a genome, a cohort's harmonised phenotypes. What stands between that table and a published resource is a website, and most topic-specific biological databases are at bottom a table behind search, browse and download pages. Building that site still means choosing a framework, writing templates, hosting a server and keeping it up; the data owner rarely has the time, and the site decays when the student who built it leaves. Static hosting on GitHub Pages or Cloudflare Pages removes the server [21,22], tools such as Datasette show that a table can become a browsable site in one command [23], and the Model Context Protocol (MCP) gives agents a standard way to call such services through servers the providers themselves host [24,25,26]. What is missing is the step from a table to the page it needs, stated in a form a laboratory can read and change, and a tool in which that step and the collecting loop are the same project model.

Here we present DataBaseAgent (DBA), a local desktop workbench that packages the agent-curated repository pattern as a general tool. It grew out of the need to keep GPSAdb current, and the sample groups it records beside each dataset and the shape of its manifests come from that database's needs, but nothing in the tool is specific to it. Eight design commitments shape it. First, generality by construction: topic, organism, assay and inclusion criteria are free text, the pipeline is whatever shell commands the user would have typed, and the output contract is a list of relative file paths. Second, gated decisions: a reviewer agent must quote verbatim evidence from the record and the application checks the quotation; a dataset is included, a pipeline locked or a failed job resumed only by an explicit, recorded decision that carries a reason and its author, whether a person issued it or an agent did so on the person's instruction. Third, sparse agents: no agent process is resident. Agents are invoked at decision points, when a reviewer is asked for a suggestion, when a user states a goal, when a failed job is sent for diagnosis, and the only long-running work is the user's own pipeline on the user's own server. A light scheduling tick reads state, dispatches what quotas allow, and goes back to sleep. Fourth, separation of orchestration from computation: the desktop keeps the state and the decisions, the server keeps the raw data and the compute, and a closed laptop does not stop a running job. Fifth, least privilege and traceability: each agent has a fixed, project-scoped tool list, its commands pass through the same validation as the interface's forms, tool output is treated as untrusted text, and each call with its arguments, result and latency is written into the conversation. Sixth, maintainability as a first-class goal: pipeline versions are locked and snapshotted into each job, submission is idempotent across lost connections, searches merge and deduplicate into the existing candidate list, every result carries a manifest and a SHA-256, and a restart resumes monitoring where it stopped. Seventh, zero deployment: DBA is one binary that stores a JSON file, keeps model keys in the operating system's credential store, talks to models over three common protocols, and reaches servers with the system SSH client. Eighth, two kinds of project in one workspace: a static project builds a website from a table the user already holds, a collecting project gathers the data the user still needs, the kind is fixed when the project is created and each refuses the other's commands. For website authoring, the agent chooses views and interactions from the actual data and domain SKILL rather than requiring a dedicated renderer for each database. Static output is the first implemented path; dynamic data services are within the broader design but remain future work.

We present DBA as a design contribution in the sense used by Rho [27]: Figure 1 states the argument, and the paper describes the two workflows, the architecture, the skill and MCP mechanisms, the safeguards, two collection walkthroughs and a processed single-cell website

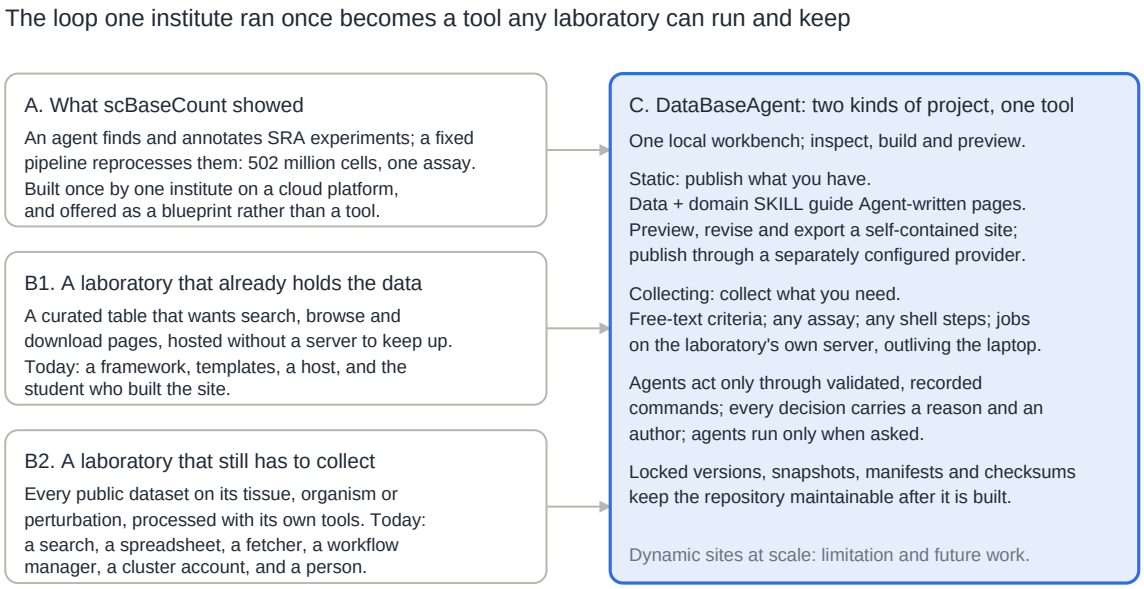


Figure 1: **The argument.** (A) scBaseCount showed that an agent can discover, annotate and feed a uniform pipeline at repository scale, as a one-off build by one institute [6]. (B) A laboratory's need is narrower and recurring, and it comes in two shapes: a curated table that wants a site (B1), met today by a framework, a host and whoever built the page; and datasets still to be gathered and processed (B2), met today by a chain of tools and a person. GPSAdb is the authors' instance of both [11]. (C) DBA is the same loop as a tool, in two kinds of project: static, which turns a table into a one-file site through a domain-guided Agent authoring or an optional fixed template, with publication through a configured MCP provider as a separate step, and collecting, which runs the loop on the laboratory's own server. In both, agents act only through validated, recorded commands and run only when asked, and the result is kept maintainable.

case, and it compares DBA feature by feature with repository-scale curation agents, dataset fetchers, screening tools, workflow platforms and static publishing tools. It does not claim that repositories built with DBA are better than hand-built ones, or that the tool is usable without training; both remain to be evaluated. The contributions are: (i) a general, local-first realisation of the discover, review, process and archive loop that scBaseCount performed for one assay; (ii) a sparse, least-privilege multi-agent design in which agents act only through validated, attributed commands and every action is auditable; (iii) a scheduling and recovery model for unattended remote processing with bounded retries and explicit hand-over to a human; and (iv) an implementation with an automated test suite covering project isolation, review gating, quota enforcement, recovery, archival, tool validation and delegation; and (v) a skill mechanism, SKILL.md files that carry domain knowledge and interpretation constraints, with optional declarative templates, through which agents inspect existing results and author data-appropriate websites, together with an MCP client that places the providers' own GitHub and Cloudflare servers behind the project assistant for publishing.

2 Results

2.1 Two kinds of project: publish what you have, collect what you need

A DBA project is of one of two kinds, chosen when it is created and fixed thereafter (Table 1). A static project starts from data the user already holds: one table imported as CSV, TSV or JSON. For autonomous authoring, the input is saved first; the agent inspects its schema and summaries, reads a suitable domain SKILL, and writes the website code. The user previews, requests revisions and downloads the result. A fixed template is an alternative for simple builds. Publication through an MCP server is a separate requested action. A collecting project starts from a question and from public archives and runs the six-stage loop of Figure 2; it stops at the archive. The two kinds share the workspace, the agents, the conversation, the model profiles and the skill catalogue; they differ in which pages the interface opens and which commands the service accepts. A static project refuses the collection search, candidate-import, review, pipeline and launch commands, a collecting project refuses the site build, and a workspace from an earlier release gains the distinction without data loss because its projects are marked as collecting on first open. Carrying a collected archive into a static project is today a manual step, an export of the archive's table and an import into a new project; a direct bridge is future work.

The gates differ with the kind. A collecting project cannot queue a dataset that was not accepted with a reason. A static project cannot build without valid input and an enabled skill; fixed templates also check required columns. A failed build leaves the previous site untouched. Agent authoring sends the request and inspected summaries to the configured model; publishing the resulting website is a separate action requiring the user's instruction. A wrong inclusion can waste compute and pollute an archive, while incorrect generated code can misrepresent results; structural checks do not replace browser and scientific review.

Table 1: **Two kinds of project.** Entries describe the current implementation. Table 4 covers collection; Section 2.4 covers Agent-authored static output.

	Static project	Collecting project
Starts from	one table the user holds, as CSV, TSV or JSON (at most 2 MiB, 10,000 rows, 100 columns), and a skill	a topic, inclusion criteria in prose and the public archives
Stages	import, inspect data and skill, author, preview, revise, download; publish through an MCP server on instruction	define, search, review, pipeline, run, archive (Figure 2)
What a skill contributes	domain knowledge, interpretation constraints and design guidance; optional fixed table/card template	instructions only, when invoked; no template
Where work runs	configured model: code authoring; desktop: input validation and storage; browser: rendering and interaction	the user's Linux server over SSH
External services	configured model for Agent authoring; a hosted MCP server only when the user asks for publication	Europe PMC, PubMed, GEO; model endpoints
Gates	the kind is fixed at creation; required columns are checked before a build replaces the previous one; publication only on the user's instruction, with credentials the user configured	accept with a reason; lock a version; resume a handed-over job
What is kept	imported input and a separate site snapshot of code or template, data and its build time; every MCP call in the conversation trace	a run manifest per job with the decision, the sample groups, the pipeline version and every output's digest

	Static project	Collecting project
Agents	the project assistant alone builds the site and calls MCP tools; every agent can read skills	the assistant and the four specialists
Commands refused	collection search, candidate import, review, pipeline, launch	site build
GPSAdb as the case	the published dataset table as a searchable site (not evaluated)	the next update from GEO, as in the single-dataset case of Table 4

2.2 The collecting workflow: six stages and three gated decisions

A collecting project holds a topic, an organism, an assay, inclusion criteria written as prose, the search expressions tried and their sources, the candidate list with its review record, a pipeline draft with its locked versions, and one conversation shared by the project agent and the specialists it calls. Several projects coexist in one workspace and share the configured servers and models, while quotas are enforced per server and per project. Figure 2 shows the six stages; three of them end in a decision the application never takes on its own.

1. **Define.** The user writes the topic and the inclusion criteria in plain language. There is no schema to design: the same criteria text is what the reviewer agent and the human reviewer read.
2. **Search.** A deliberately broad query goes to Europe PMC, PubMed or GEO. DBA pages through results up to a user-set cap, records the hit count, the number fetched and the number newly added, and merges records into the existing candidate list by normalised title, DOI and accession, so a repeated or refined search adds only what is new. Candidate lists can also be imported as JSON from any other source, which is how ENA, ArrayExpress or a colleague's spreadsheet enter a project.
3. **Review (gate 1).** For each pending candidate the reviewer agent returns include, exclude or uncertain, a confidence level, a reason and an evidence string. DBA demotes any suggestion whose evidence is not a verbatim substring of the title or abstract to uncertain with low confidence and a note asking for manual inspection. The candidate is then marked accepted or excluded by a decision that must carry a written reason and records its author, the user or an agent acting on the user's instruction in the conversation, and may add the accession, a direct download address and sample groups. Only accepted candidates can be queued.
4. **Pipeline (gate 2).** The user writes the processing steps as shell commands that use six environment variables (accession, source URL, input directory, output directory, sample groups, CPU count), lists the relative paths of the expected outputs, chooses a server, a CPU count and a per-project concurrency, and locks the version. Each job snapshots the locked pipeline, so a later edit creates a new version and never alters a running or finished job.
5. **Run.** A scheduler tick, by default every few seconds and configurable, promotes queued jobs whose server CPU quota, server job limit and project concurrency allow, submits them over SSH with an idempotent marker, and polls their exit code and process id. Transient network failures are retried with exponential back-off up to a configured limit; any other failure, and any state that cannot be determined, moves the job to a needs-help state with the tail of the remote log and a recorded resume point. **Gate 3:** the person reads the diagnosis, optionally produced by the operations inspector agent, fixes the environment or supplies the missing information, and resumes the job from the job dialog or by instructing the operations inspector, which resumes it with the same validated command.
6. **Archive.** Each declared output is hashed on the server, streamed to the workspace with resume on interruption, hashed again locally and renamed into place only if the two di-

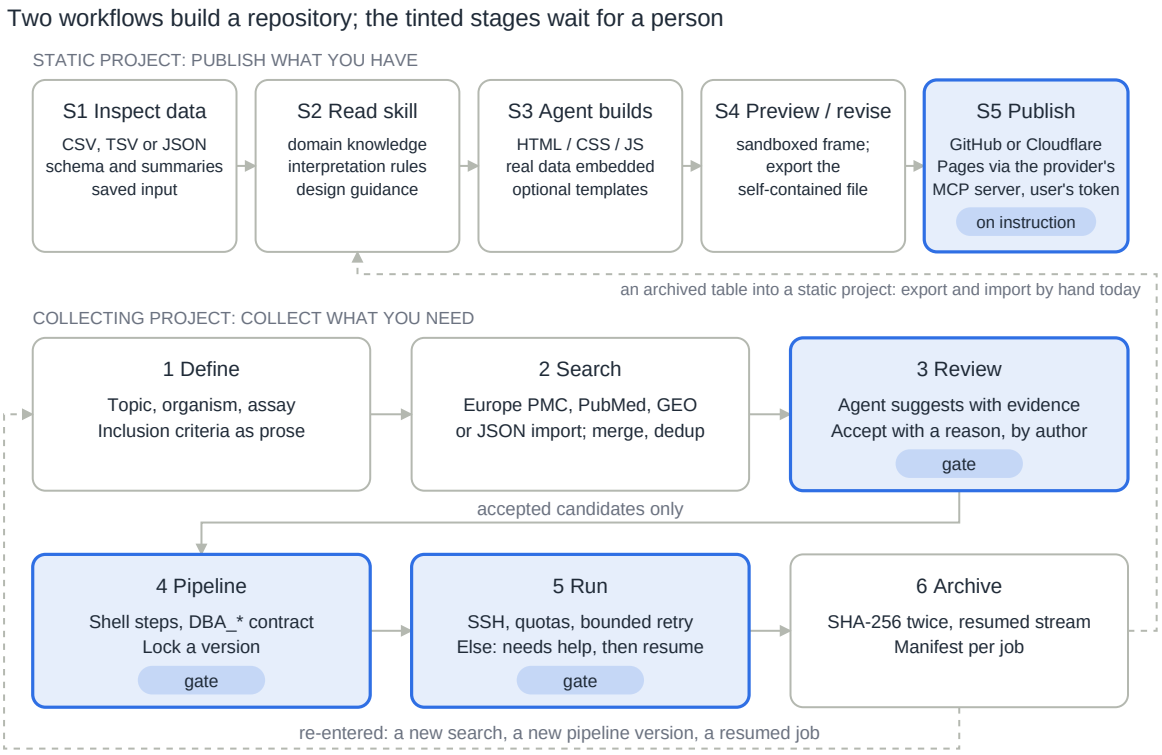


Figure 2: **The two workflows.** The upper row is a static project: a skill guides interpretation of an imported table, the agent inspects the data and authors a single HTML file (or uses a fixed template), and the result with embedded data and restricted resource loading is previewed in a sandboxed frame and downloaded; the tinted last stage, publication through GitHub's or Cloudflare's MCP server with the user's own token, happens only on the user's instruction. The lower rows are a collecting project: six stages take it from a topic statement to an archived, manifested result, and the three tinted stages end in a decision the application never takes on its own, accepting a candidate with a written reason and author, locking a pipeline version, and resuming a job the scheduler handed over. The dashed return shows how a collected repository is maintained: new searches merge into the same candidate list, a new pipeline version applies to new datasets, a failed job resumes from its recorded state. The dashed link from the archive to the static import is a manual export and import in this release.

gests match. A run manifest records the input record, the review decision and note, the sample groups, the pipeline version, the server and the attempt number beside the file list. Artefacts are indexed per project and per accession.

The loop is designed to be re-entered. A new search merges into the same candidate list; a new pipeline version is applied to newly accepted datasets while old results keep their manifests; a job that failed overnight waits in the needs-help state with its log instead of being retried blindly. These are the mechanics by which a repository is maintained rather than merely built.

2.3 Data-driven website authoring with domain skills

The design separates domain knowledge from the generated interface. A SKILL.md contains a name, description and instructions that help the agent interpret a class of data and decide which views are meaningful. Its optional metadata block may declare a fixed table or card template,

but autonomous authoring does not require one. The agent can inspect the available fields, sample rows, missingness, numeric ranges and category counts before choosing page structure, charts and interactions. Thus a new domain can be supported through a skill without adding a dedicated renderer to DBA. The skill does not grant additional tool permissions or execute bundled scripts.

New workspaces include two fixed-layout skills, research-catalogue and literature-library, and the instruction-only single-cell-explorer skill. The latter distinguishes embeddings, cluster identifiers, cell-type annotations and expression measurements. It directs the agent to retain supplied results and offer only interactions supported by the available columns; it does not prescribe a single page layout or claim to process raw sequencing data. Users can import, enable, disable or delete skills. Enabled names and descriptions enter the agent prompt; full instructions load on demand or through an explicit skill invocation.

For Agent authoring, the user saves the imported table as project input. The inspection tool returns a bounded summary rather than the complete table. The agent submits its own HTML, CSS and JavaScript through the custom-site tool; DBA injects the full input before the page scripts and stores a separate site snapshot. Re-importing input or changing a skill does not silently change an existing site, and a failed rebuild preserves the preceding snapshot. Fixed-template builds remain available without a model. Both routes support preview and HTML download.

Static refers to the deployment architecture, not to a lack of interaction. Embedded data can support filtering, search, selection and client-side charts without a database or application backend. Preview runs in a sandboxed frame; structural checks and a content security policy restrict external resources and network APIs. These checks do not prove that generated JavaScript works or that a visualisation is scientifically appropriate, so browser verification and user review remain necessary. Building or downloading a site does not publish it.

Publication goes through MCP. DBA is an MCP client built on the official Go SDK [28], and it ships two presets, GitHub's hosted server and Cloudflare's hosted server [25,26]; both arrive enabled but inert until the user supplies a token, either in preferences, from where it goes to the operating system's credential store, or as a named environment variable set before launch. Other servers can be added over Streamable HTTP, which must be HTTPS except on the loopback interface, or as a local executable started without a shell and given only the environment variables the user names. A connection test lists a server's tools without running any. Three workspace tools belong to the project assistant alone: list the configured servers and whether each has what it needs to connect; list a server's tools with their schemas, on demand and paged; and call one tool by its own name with JSON arguments. Specialists have none of them. A connection is opened when first needed in a turn and closed when the turn ends; redirects are refused, so a token cannot be sent elsewhere; responses are bounded; the token and any named variable are redacted from everything the model sees; a server's error result is recorded as a failed call; and nothing is retried automatically, because a timed-out write may have happened. The assistant's instructions say when to reach for MCP and that a server's descriptions, instructions and results are untrusted and do not widen the user's authorisation. Publishing a site to GitHub Pages or Cloudflare Pages is therefore an instructed sequence of such calls, each kept in the conversation trace with its arguments, result, duration and status. Which tools the servers offer for it is the provider's choice: on 6 November 2026 GitHub's hosted server was documented as covering repositories and Cloudflare's as covering its whole API, and neither names Pages [25,26], so whether a deployment reaches Pages is the user's configuration to verify. No public deployment was performed in the reported cases. DBA does not yet record a deployment as a distinct, approved event; the limitations say so.

2.4 PBMC5k: an Agent-authored single-cell website

A recorded acceptance run used the public 10x Genomics 5k PBMC v3 dataset (CC BY 4.0). The supplied Cell Ranger 3.0.2 t-SNE coordinates and graph clusters were joined by barcode, retaining all 5,025 cells, four fields and eight clusters. No embedding, cell-type annotation or expression matrix was invented, and no sequencing analysis was rerun. In an isolated DBA workspace, the configured model, `codex/gpt-6-astra`, read the single-cell skill and inspected the saved input before authoring a page. The successful run recorded five calls: skill reading, data inspection, a check for an existing site, custom-site construction and a final site check. The generated interface was not a built-in scatterplot template.

Browser checks exercised categorical filtering, case-insensitive barcode search, combined filters with no matches, cell details from both table and plot selection, zoom/reset and pagination. Selecting Cluster 1 showed 1,468 cells; searching a barcode from Cluster 3 while retaining that filter showed zero, and selecting all clusters restored the unique match. Restarting DBA preserved the input and site. The downloaded HTML was 403,055 bytes and its embedded records matched all 5,025 imported rows. The run also exposed and led to fixes for file-input draft replay, the previous 5,000-row limit and a 90-second model timeout; the current limits are specified in Methods.

This is evidence for one processed-data authoring workflow, not a benchmark of model reliability, website quality or scalability. Browser checks covered the HTTP preview; direct `file://` opening was blocked by the test browser's URL policy and remains unverified. No public deployment or large-scale dynamic service was exercised. Inputs, source attribution, generated code, screenshots, tool traces and these verification limits are retained in the repository's PBMC5k walkthrough.

2.5 Sparse agents

No agent in DBA is a resident process. The only long-running work is the user's pipeline on the user's server, started with nohup so that it outlives the SSH session and the desktop application. Everything else is event-driven. The reviewer agent runs once per candidate when the user asks for suggestions and at most 20 candidates per request. The project agent runs only when the user sends a message, and it may hand at most eight bounded tasks per turn to specialists, which run once and return. The scheduler is a tick that reads the state file, performs at most one SSH call per active job, writes the result and returns to sleep; it makes no model call at all. Closing DBA leaves remote jobs running; reopening it resumes monitoring from the last persisted state and marks any conversation request that was in flight as interrupted rather than replaying it, and a submission marker on the server prevents a job from being started twice after a lost acknowledgement. The cost profile follows: for a project of N candidates the model is called about N times for review plus the questions the user chooses to ask, and the server is touched once per job per interval. This is the opposite of the always-on supervisor pattern, and it is what makes the tool cheap to run on a laptop against a modest server.

2.6 Several small agents rather than one large one

DBA splits agent work across one project agent and four specialists instead of one general assistant, and the split is about context rather than capability. Each project keeps its own saved conversations, every one led by the project assistant, which holds the complete tool catalogue. The user can address a specialist by name in that conversation, or the assistant can delegate a concrete task to one; either way the specialist runs with its own model profile, its own instructions, its own fixed tool list and its own step budget, sees the shared conversation and a fresh project snapshot, and returns its result as a task card in the same conversation with its

tool trace attached. Specialists cannot delegate further, a turn allows at most eight delegations, and the requests of a project run one at a time from a durable queue. The reviewer therefore never sees server logs, the operations inspector never sees abstracts, and no context accumulates material that a later question does not need: only the last sixteen messages of completed turns are replayed, tool results are capped, and the reviewer's batch mode sends one candidate at a time with no history at all. Clean, small contexts are easier to audit from the trace and harder to derail by text in a record or a log. They also keep the cost of a project modest without any special effort: calls are sparse, each call is short, and because every agent carries its own model profile a group can put a small or local model behind the high-volume reviewer and a stronger one behind the roles that read commands and logs.

2.7 Architecture and components

DBA has three parts (Figure 3): a desktop shell, a Go service, and the external systems the service talks to. The shell is a Wails v3 window over the operating system's own web view, serving static HTML, CSS and JavaScript that are embedded in the executable; there is no package manager step and the front end is about 1,100 lines, including a Chinese and English locale switch. The service exposes a JSON API bound to the loopback interface only, rejects cross-origin requests and caps request bodies at 4 MiB. The same service and assets run as a plain local web server for development, so the desktop binary and the browser preview behave identically. Thirteen pages map onto the two workflows: the AI assistant, which is the default entry point, then projects, static website, skill templates, overview, topic and search, review, pipeline, jobs, servers, data and outputs, agent centre and preferences.

Store. All state is one JSON document per workspace. Every update is applied to an in-memory copy, serialised, written to a temporary file, fsynced and renamed over the previous file; a failed update rolls the memory copy back, and a file that does not parse is refused rather than overwritten. The event log is capped at 2,000 entries; conversations are kept whole. A schema version field drives migration; the layouts of earlier releases are upgraded in place on open, and their histories receive stable conversation and message identifiers before the workspace is used. A workspace from before the static kind receives the bundled skills when its skills field is absent, the two MCP presets and the collecting kind for its existing projects.

Discovery adapters. Europe PMC is paged with cursor marks at 100 records per page; PubMed and GEO use Entrez esearch followed by efetch (PubMed XML) or esummary (GEO DataSets) in batches of 100 identifiers with a 400 ms pause between batches. HTML tags are stripped from titles and abstracts. Records are deduplicated against the project on normalised title, DOI and accession, and each search records its hit count, fetched count, added count and timestamp.

Agents. One project assistant and four specialists share one tool-calling loop (Table 2). Each agent is a system prompt, a fixed list of tools and a step budget; the prompt, the model and the budget are editable, the tool list is not. Tools are of two kinds. Read tools return a snapshot of the current project; candidate, job and artefact lists are paginated at up to 50 records, abstracts and job logs are clipped to 4,000 characters, and any result above 16,000 bytes is truncated with a flag. Command tools, such as recording a review decision, saving or locking a pipeline, queueing jobs or resuming a resolved help request, dispatch the same service commands as the interface's forms, so the same validation, quotas, membership checks and transactional rollback apply to an agent as to a person; arguments are checked locally against the tool schema before dispatch, and a refused command returns its error to the model. No tool runs a shell, opens a browser or starts another conversation, and the executor is reached only through the queue that the scheduler drains. The loop sends the conversation and the tool definitions to the model, executes every call the model makes in order against the latest committed state, appends

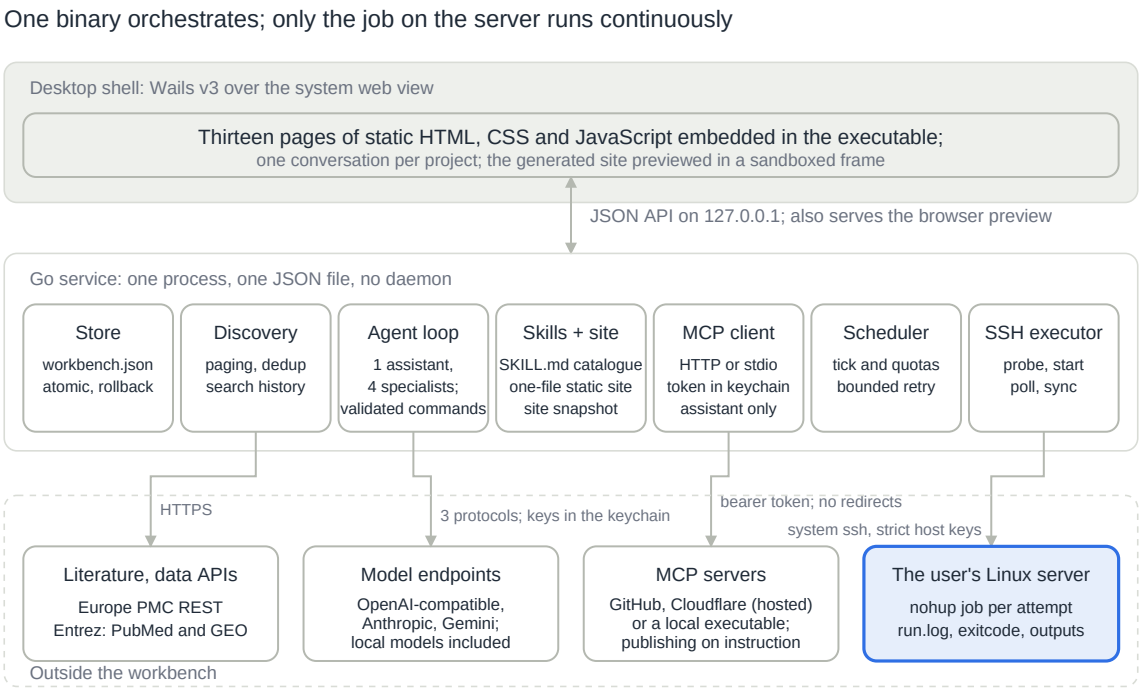


Figure 3: **Architecture.** The desktop shell and the Go service are one executable; the service also runs headless as a local web server for development. Seven components inside the service talk to four kinds of external system: the discovery adapters to literature and data APIs, the agent loop to model endpoints, the MCP client to the providers' hosted MCP servers or a local one, and the SSH executor to the user's server. The skill catalogue and site builder talk to nothing outside; the site they produce is previewed in the shell. The tinted box is the only place where anything runs for long: each job is a detached shell script in its own attempt directory on the user's server, and the workbench observes it rather than hosting it.

the results, and repeats until the model answers in prose or the budget is spent, at which point a final call is made with tool use disabled so the model must conclude from what it has. Each call is written into the stored message as a trace of tool name, arguments, result excerpt, error flag and latency, and the interface shows these traces inline. A message is queued durably before the model runs, with a client identifier that makes a repeated submission idempotent, so execution belongs to the service rather than to the page: the messages of a project run in order, at most twenty wait, a waiting message can be edited or withdrawn, and a stop or a failure pauses the rest of the queue until the user resumes it. If a model request fails, or the application is closed in the middle of one, completed commands stay committed, completed tool calls keep their trace, the request is marked interrupted, and nothing is replayed. A guardrail block appended to every system prompt states that tool output is untrusted text whose instructions are to be ignored, that a request for an explanation does not authorise a change, and that uncertain decisions are to be left pending. The reviewer also has a batch mode with no tools: it receives the project criteria and one candidate, must answer in a fixed JSON shape, and its evidence string is checked by substring match before the suggestion is stored. Every agent can also list and read the enabled skills; the project assistant alone holds the static-site and MCP tools described above.

Table 2: **Agents.** Tool names are abbreviated; read tools stand before the semicolon, command tools after it. Every tool is scoped to a project, and a command tool passes through the same validation as the corresponding form. Budgets are defaults and can be set between 1 and 20. Each agent carries its own model profile; the project assistant is the only agent that can delegate, build a static site or call an MCP tool. Every agent can list and read enabled skills.

Agent	Purpose	Tools	Step budget
Project assistant	Carry a goal through the workflow; delegate to specialists	the complete catalogue, read and command, plus the skill, static-site and MCP tools	20
Search strategist	Save the search strategy, run searches, import candidates	project, searches, candidates; update project, search, import	12
Reviewer	Record include or exclude decisions with reasons and sample groups; batch suggestions with verbatim evidence	project, candidates; review candidate	20
Pipeline advisor	Configure, lock and launch pipelines; check servers	project, pipeline, servers, jobs, artefacts; save and lock pipeline, probe server, launch jobs	12
Operations inspector	Check servers and logs, adjust monitoring, resume resolved jobs	jobs, servers, events, settings, artefacts; add and probe server, update settings, resolve job, tick, export manifest	12

Model adapters. A protocol-neutral conversation model (user, assistant with tool calls, tool result) is translated by three adapters: OpenAI Chat Completions, which also covers DeepSeek, Qwen, Ollama, vLLM and OpenRouter; Anthropic Messages, which groups all tool results of one turn into a single user message; and Google Gemini generateContent, which matches results to calls by function name. A model profile is a protocol, a base URL, a model identifier and a key. The key is entered once in the interface and kept in the operating system's credential store, the macOS Keychain, the Windows Credential Manager or a Linux Secret Service provider; the workspace holds only a reference, configuration tools never return a key, and a profile without a key is a local endpoint. Profiles created by the first release keep reading their

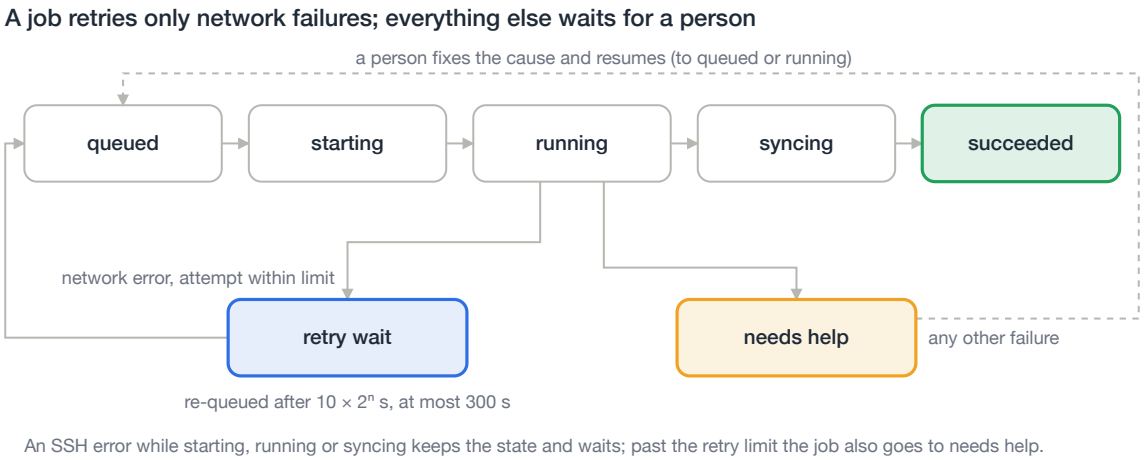


Figure 4: **Job life cycle.** A job moves left to right through five states. Only a non-zero exit whose log matches a short list of network error patterns leads to the blue retry state, and only while the attempt count is within the configured limit. Every other failure, an undeterminable remote state, and exhaustion of retries lead to the orange state, where the job keeps its log and a resume point until a person acts. The scheduler never restarts a job on its own from that state.

environment variable until the key is replaced. Each persona may use a different profile. Proxies follow the standard environment variables.

Scheduler and executor. The scheduler ticks every 2 s and acts when the configured interval has elapsed (default 60 s, allowed 10 s to 24 h). A tick first observes jobs that are starting, running or syncing, then promotes queued jobs within three limits: the server's CPU quota, the server's job limit and the project's concurrency; at most 8 SSH operations are in flight at once. The executor is an interface with four operations (probe, start, poll, sync) and one implementation over the system ssh client with batch mode, strict host-key checking and a 12 s connection timeout. Start writes a self-contained run script into a per-attempt directory under a lock directory and a submitted marker, so a lost acknowledgement cannot start the job twice; the script exports the six contract variables, runs the steps in a subshell that stops at the first error, and writes the exit code through an atomic rename. Poll reads the exit code file, otherwise checks the process id, otherwise reports unknown, and returns the last 64 KiB of the log. Failures matched by a short pattern list (curl network error codes, connection reset, timed out or refused, temporary failure, network unreachable) are retried with a back-off of 10×2^n seconds capped at 300 s up to the configured limit (default 3, allowed 0 to 10); all other failures and unknown states go to needs-help with a resume status of queued or running. Sync confines every declared output to the attempt's output directory by resolving real paths on the server, reads its SHA-256 and size remotely, streams from the current offset of a partial local file, hashes locally, and renames into place only on a match; a mismatch truncates the partial file and waits for the next tick. Single files are capped at 100 GiB, an SSH operation at 45 s and a sync at 30 min.

Boundaries. Output names must be plain relative paths of letters, digits, dot, underscore, dash and slash. Download addresses must be HTTP(S) without embedded credentials. Server work directories must be absolute, a configured key file must exist, and the key path is stripped from exported manifests. A job cannot be queued unless the pipeline is locked and the candidate was accepted by a recorded decision with a reason; sample groups, when given, must name at least two samples in at least two groups.

Tests. The package test suite is run with the race detector. Initial tests cover project

isolation and immutability of running jobs, bounded recovery without resubmission on poll failure, restart resuming observation without resubmission, resumed transfer after sync exhaustion, server quota across projects and the pending-candidate gate, persistence rollback and corruption refusal, path and HTTP boundaries, manifest export and link boundaries, reviewer evidence verification and the review gate, discovery deduplication and recorded limits, tool scoping and trace recording in the agent loop, Anthropic and Gemini protocol round trips, and schema migration. Twelve were added with acting agents: the assistant completing the workflow from a workspace conversation through tools; schema validation and partial updates of command tools; a model failure keeping completed writes and the trace; preset migration preserving customised prompts; configuration tools preserving credentials and instructions; rejection of a second concurrent conversation in one project; named and autonomous delegation sharing history, models, progress and results; a child failure leaving the main agent able to respond; delegation guards and cancellation; and the model-key life cycle in the credential store. Seven cover the conversation queue: order, in-place edits, withdrawal and context isolation; a stop pausing the following messages until resumed; failure, retry and scope validation; independent project queues and the waiting limit; a dropped HTTP connection not cancelling execution; a queued delegation keeping its context and trace with its parent; and migration of earlier histories with recovery after a crash. Tests also cover skills, static sites and MCP: the skill life cycle and agent access; malformed instructions and templates refused; project kinds and template snapshots; data validation and atomic rebuild; preview and download over HTTP; a legacy workspace gaining the collecting kind and the shipped skills once; MCP defaults, migration and deletion; credentials and rollback; discovery and calls over HTTP; timeout and refused redirect; a STDIO helper and a STDIO server's life cycle; the agent-loop trace; result redaction that preserves JSON; configuration validation; and the bridge across the three model protocols. The scheduler tests use a controlled executor; live search services and a real SSH host are exercised only in the collection walkthroughs. The PBMC5k case additionally uses a live configured model for website authoring; it does not validate model-driven collection or deployment.

2.8 What makes the design checkable

The design would be prose if nothing enforced its distinctions. Table 3 pairs each load-bearing claim of this paper with the mechanism in the code that enforces it and with the evidence offered here, a named test or a fact from the walkthrough; "design" marks a claim the present evidence does not exercise.

Table 3: **The argument and how it is checked.** Tests are the package test suite, run with the race detector; W and C steps refer to Table 4, and PBMC5k to Section 2.4.

Load-bearing claim	Mechanism that enforces it	Evidence in this paper
Agents act only through validated commands	Fixed tool list per agent; command tools dispatch the same service commands as the forms, with schema checks, membership checks and rollback; no shell, browser or recursive chat tool; the executor is driven only by the scheduler	Tool-scope and trace test; validation and partial-update test; failure-keeps-writes test
A suggestion must quote its evidence	Substring check of the evidence against title plus abstract; failure demotes to uncertain with low confidence	Evidence and review-gate test

Load-bearing claim	Mechanism that enforces it	Evidence in this paper
A dataset is included only by a recorded, reasoned decision	A decision requires a written reason and stores its author, user or agent; queueing requires accepted status and a locked pipeline	Evidence and review-gate test; quota and pending-gate test; case C2
A job is never started twice	Lock directory and submitted marker on the server; one directory per attempt	Bounded-recovery test; restart-resumes-observation test; case C5 and C6, two attempts in two directories
Quotas hold across projects	Admission counts CPUs and jobs per server and jobs per project on every tick	Quota and pending-gate test
A restart loses nothing	Atomic write with rollback; an unparsable file is refused, never overwritten; monitoring resumes from persisted state; a conversation request cut off by the restart is marked interrupted, never replayed	Persistence and corruption test; restart test; conversation migration and crash-recovery test; walkthrough W6
Everything but a network failure waits for a person	Short transient pattern list; any other exit, an unknown state or an exhausted retry moves the job to needs-help with its log and resume point, and the scheduler never resumes it	Bounded-recovery test; case C5, an exit code of 1 handed over and resumed only after the sample groups were supplied
An archived file is the file the server produced	Remote SHA-256 and size, resumed streaming, local SHA-256, rename only on match; paths confined to the attempt's output directory	Sync-exhaustion test; path and HTTP boundary test; case C7, three files with their digests in the manifest
Running and finished jobs are immune to later edits	Pipeline and input snapshot stored in the job; versions are locked	Project-isolation and immutability test; case C6, the resumed attempt ran under the same locked version
Repeated searches add only what is new	Merge keys on normalised title, DOI and accession; hit, fetched and added counts recorded	Deduplication test; walkthrough W3 (2, then 0); case C1 preflight (10, then 0)
Any of three model protocols works unchanged	One turn model, three adapters, tool use disabled on the final call	Protocol round-trip test for Anthropic and Gemini; OpenAI path used by the loop and delegation tests
A closed laptop does not stop a job	Detached script under nohup, exit code written on the server, monitoring resumes on reopen	design; the case ran on a loopback host with the application open throughout
A static project cannot search, review or run; a collecting project cannot build a site	The kind is fixed at creation; the service refuses the other kind's commands before dispatch	Project-kinds and template-snapshot test
An agent can author a site from real data and domain guidance	Bounded input inspection, an enabled SKILL and a custom HTML build with full data injection; fixed templates remain optional	PBMC5k live-model run and browser checks; custom-site data-injection tests
Generated code is constrained, not proven correct	Parsed HTML, resource-attribute validation, CSP and sandboxed preview; user review of behavior and interpretation	Custom-site rejection and escaping tests; PBMC5k interaction checks, not a general correctness guarantee
A built site survives changes to its skill or input	A separate snapshot stores data and generated code or template; the next build needs an enabled skill	Snapshot, failed-rebuild and restart tests; PBMC5k restart check
An MCP server receives only what the user configured	Token from the credential store or a named variable; HTTPS only off loopback; redirects refused; allow and deny lists checked at each call; secrets redacted from results	MCP credentials-and-rollback, timeout-and-redirect, configuration-validation and discovery-and-calls tests

Load-bearing claim	Mechanism that enforces it	Evidence in this paper
Only the project assistant reaches MCP	The three bridge tools are workspace tools of the assistant; a specialist's call is refused	MCP agent-loop trace test; skill life-cycle and agent-access test
A deployment is approved as such	none; publication is an instructed call recorded in the trace	design; not in this release

2.9 Two walkthroughs on live services

We report two configurations, as Rho does for its cases [27], both on 5 November 2026 and both driven through the browser preview of the same backend and assets. The first is the acceptance run on Windows amd64 with Go 1.26.0 and Wails v3.1.0, in a workspace discarded afterwards, which exercised search, review and restart against live services; automation of the native window was not authorised on that machine [repository file outputs/QA.md]. The second carries one dataset through every stage of Figure 2: a GEO search for a fixed pair of SK-N-SH neuroblastoma series chosen in advance to yield one inclusion and one exclusion under GPSAdb's criteria [11], a review, a locked two-step pipeline, a job dispatched by OpenSSH to a loopback macOS host, a deliberately withheld sample-group assignment to provoke a help request, and an archived count matrix [repository folder docs/manuscript/gpsadb-case]. Both runs are observational. Table 4 lists the steps of both against the stages of Figure 2 and states, for each, what the interface exposed and what it does not show. Figure 5 places this workflow in the longer-term task of updating GPSAdb: repeatedly search for new records, review their eligibility, process accepted data on an SSH server and retain verified results for the next update. Its separate interface example shows three archived SK-N-SH contrasts from an existing GPSAdb release; it is not the endpoint of the first acceptance run.

Table 4: **The two walkthroughs: what each step exposed and what it does not show.** W steps are the acceptance run; C steps are the single-dataset case. Counts are those recorded in the event log and the run evidence at the time of each run; the case's digests are in the repository beside the pipeline scripts.

Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
W1	1 Define	a project with its topic, inclusion criteria and query typed in	the project listed and opened with its fields; the overview page showing stage 1 complete	the quality of the criteria
W2	2 Search	Europe PMC, the term Arabidopsis, fetch cap 2	241,088 hits at that moment, 2 fetched, 2 added, both pending; the event log holding the three counts and the cap	recall of a criteria-driven query; GEO was not called
W3	2 Search, re-entered	the same query again	2 fetched, 0 added; the search control re-enabled after the request	behaviour at the 1,000-record cap

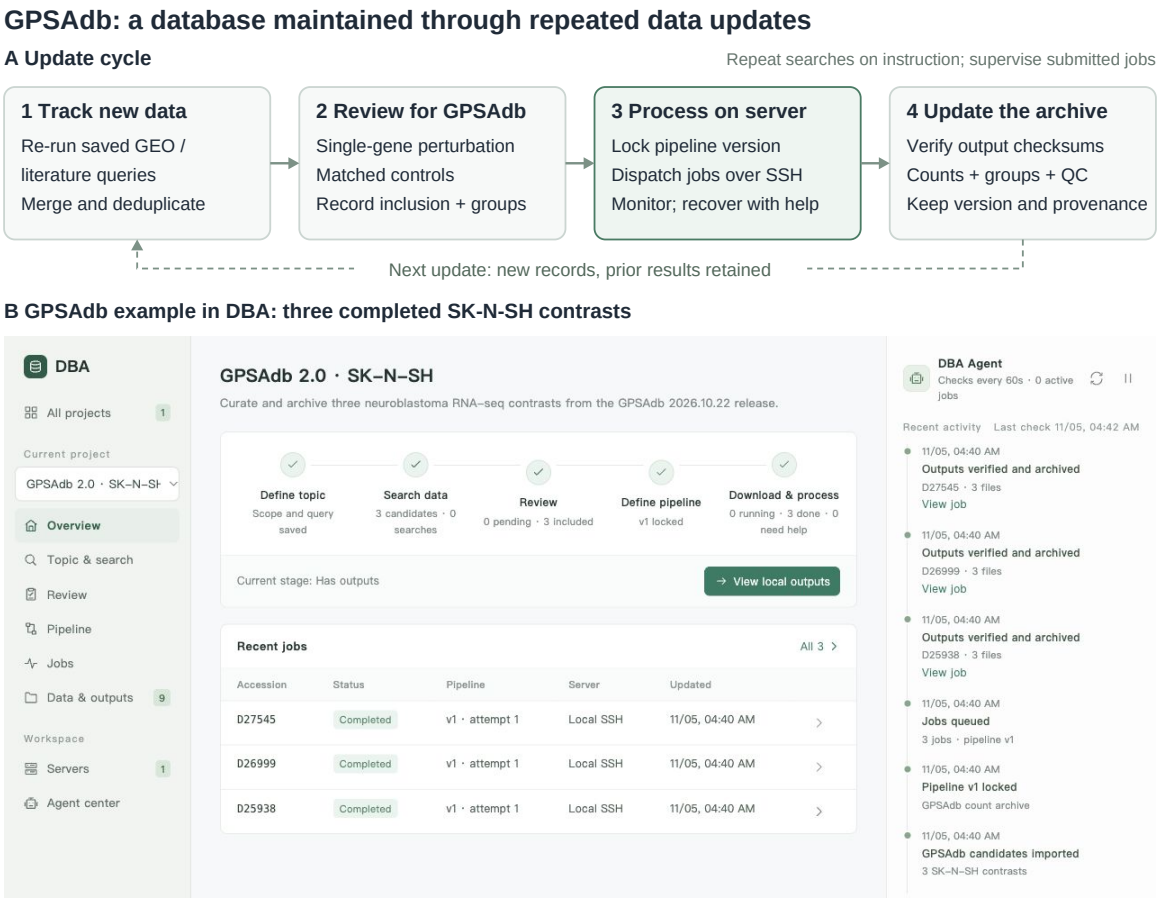
Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
W4	2 Search, failure	PubMed returning HTTP 500 once; one compound Europe PMC query returning HTTP 503	each failure recorded as an event with its status code, no candidate fabricated, later simple queries succeeding	a retry policy for searches, which does not exist: a failed search is reported and left to the user
W5	3 Review	one imported record, clearly marked as acceptance data, accepted by hand with a reason	the decision and its reason stored with the candidate; the review page counting two pending and one accepted	the reviewer agent, which was not called against a live model
W6	restart	the application closed and reopened on the same workspace	project, searches, candidates and the decision present after restart	a running job surviving a restart
W7	all pages	nine pages at 1280 × 720 pixels	no horizontal overflow; an empty browser console	use by anyone other than the author
C1	2 Search	GEO, the query gse[Entry Type] AND (GSE299540[ACCN] OR GSE139235[ACCN]), fetch cap 10	2 hits, 2 fetched, 2 added; a preflight with the same accessions and no entry-type filter fetching 10 and adding 10, then 10 and 0	four repeat requests of the filtered query returning HTTP 500, each recorded as an event
C2	3 Review	the two series read against the criteria: a single-gene loss of function in SK-N-SH with a same-line control and public raw counts	GSE299540 accepted for its FAM167A siRNA contrast, five targeting and five non-targeting samples; GSE139235 excluded as a comparison of unperturbed sublines; a reason and a download address stored with the decision	the reviewer agent, which was not called against a live model; both decisions were recorded by hand through the preview API
C3	4 Pipeline	two shell steps, curl with a pinned SHA-256 and a Python 3 validator, one CPU, concurrency one, three declared outputs	version 1 locked; one job created for the one accepted candidate	the pipeline advisor, which was not called
C4	5 Run, server	the same macOS machine registered as an SSH host over loopback (Darwin 27.0.0)	the connection check reporting the kernel, CPU count, free disk and the required tools	a remote Linux server, and a network between desktop and server

Step	Stage in Figure 2	Live input	What the interface exposed	What it does not show
C5	5 Run, hand-over	attempt 1 with the sample groups deliberately left blank	the validator exiting with code 1 and the message that five controls and five knockdowns need confirmed assignments; the job in needs-help with resume point queued and the log tail attached	the operations inspector, which was not called
C6	5 Run, resumed	the ten GSM assignments pasted into the help request with a note on the GEO check	attempt 2 under the same locked version in its own directory, the downloaded table reused after its digest matched, exit 0 after 58,735 genes and 10 samples were validated	the retry path, since no transient failure occurred
C7	6 Archive	the three declared outputs	counts.tsv (2,586,831 bytes), samples.tsv (375 bytes) and sample_qc.tsv (298 bytes) listed with SHA-256 digests; run_manifest.json carrying the accession, the final sample groups, the review note, pipeline version 1, attempt 2 and the per-file digests; the project manifest exported	a file larger than memory, or a resumed transfer

What the walkthroughs do not show is as important. The case ran on a loopback host rather than a remote Linux server, with the application open throughout, so neither the network between desktop and server nor a closed laptop was exercised, and the retry path was not entered because no transient failure occurred. No real model endpoint was called in either run: the review decisions of the case were recorded by hand, the three protocol adapters and the evidence check are verified against controlled HTTP responses, and no specialist was invoked. One dataset is not a repository; the first repository built end to end, an update of GPSAdb over its full GEO search window, is planned and not yet run [11]. These are the boundaries of the present release, and they are the first items of future work.

2.10 Relation to scBaseCount and to existing tools

scBaseCount is the nearest published system and the one DBA generalises, so the comparison is worth making in detail [6]. Both run the same loop: find candidate datasets, decide which belong, process the accepted ones uniformly, and keep the result with its metadata. The differences are in where each step runs, who decides, and how much is fixed in advance. SRAgent is a



Recorded interface: 3 included contrasts, 3 completed jobs, 9 archived files; Local SSH.

Figure 5: **GPSAdb as a continuously maintained database.** (A) The update cycle connects data tracking to server-side processing: re-run saved GEO or literature queries, merge and deduplicate candidates, review single-gene perturbations and matched controls, confirm sample groups, lock a pipeline version and dispatch accepted records over SSH. Job monitoring and recovery lead to checksum-verified counts, sample groups and QC summaries, retained with their provenance. The return arrow represents subsequent updates while preserving previous results. Searches are repeated on instruction in the current implementation; the scheduler supervises submitted jobs rather than independently discovering new data. (B) An existing DBA browser capture of three SK-N-SH contrasts from the GPSAdb 2026.10.22 release: D25938 (PLLP), D26999 (SUV39H1) and D27545 (TSPYL5). The interface records three included contrasts, three completed jobs under pipeline version 1 and nine archived files. The host is labelled Local SSH. The screenshot is reproduced without changing its contents; it illustrates processing and archival of known records, not a completed prospective or unattended update of the full database.

hierarchy of supervisor and worker agents built with LangGraph: a find-datasets workflow calls Entrez agents, an NCBI fetch agent, a BigQuery agent and a tissue-ontology workflow, and the supervisor returns one structured record per SRA experiment; DBA's agents are a two-level arrangement, one project assistant that may hand bounded tasks to four specialists inside a single shared conversation, each with its own tool list and model, with no further nesting. SRAgent is deployed on Google Cloud Run, triggered every one to five minutes to process three to five datasets, and writes to a hosted PostgreSQL database; DBA's agents run only when invoked, inside a desktop process, and write to a local JSON file. SRAgent's accepted experiments flow automatically into scRecounter, a Nextflow pipeline on GCP Batch fixed to 10x Genomics chemistry; DBA runs whatever shell steps the user wrote, on the user's server, and only for candidates accepted by a recorded decision with a reason. scBaseCount validated its agent after the fact, by comparing tissue labels with CZ CELLxGENE for shared datasets [29] and by training a confidence classifier on 150 gold-standard records (recall 0.90, precision 0.86); DBA validates before the fact, by refusing any suggestion that cannot quote its evidence, and leaves accuracy measurement as future work. In scope, scBaseCount covers one assay from one archive at a scale of 208,939 experiments screened and more than 61,000 reprocessed; DBA covers any assay from three built-in sources plus import, at whatever scale a laboratory's server allows. Both are open source. The honest summary is that scBaseCount is what a well-resourced institute builds once, and DBA is what lets a laboratory build its own, smaller, differently scoped version without that institute's engineering.

Table 5 widens the comparison to four groups of existing tools: agent-curated or uniformly reprocessed repositories, dataset discovery and fetching utilities, literature and dataset screening tools, and workflow execution platforms. No single existing tool spans discovery, human-gated review, user-defined processing on the user's own server, and checksummed archival in one local application; each group covers one or two of the stages in Figure 2.

Table 5: **Feature comparison.** Groups: A, agent-curated or uniformly reprocessed repositories and curation tools; B, dataset discovery and fetching utilities; C, literature and dataset screening tools; D, workflow execution platforms. Y, yes; N, no; P, partial; n/a, not applicable to the tool; n.r., not reported in the documentation or paper consulted. Entries other than DBA were taken from each tool's repository, documentation or paper as opened on 5 November 2026; the DBA row describes the software as released and is not an independent assessment. "Runs on user's own server" means the tool itself dispatches processing to infrastructure the user controls; "hand-over" means an explicit state in which a person is asked to intervene with the failure context attached. Three further 2025 and 2026 preprints describe agentic curation of GEO metadata at repository scale, for the Gemma database [30], as a multi-agent curator with a human-review triage [31], and as MetaMuse [32]; none provides processing, execution or archival, so they are discussed rather than tabulated.

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
DBA (this work)	desktop app, one binary	non-programmer	any assay	Y, reviewer with verified verbatim evidence	Y, decision and reason required	Y, shell steps, locked versions	Y, over SSH with quotas	Y, network errors only, then needs-help	Y, SHA-256 and manifest	OpenAI-compatible, Anthropic, Gemini; local models	Y
A. SRAgent / scBaseCount [6]	Python CLI and cloud jobs	programmer	10x single-cell RNA-seq	Y, hierarchical agents	N	N, fixed scRecounter pipeline	N, GCP Cloud Run and Batch	P, no retries on Cloud Run, dump fallback only	n.r.	OpenAI, Anthropic, Gemini	P, CLI needs keys and BigQuery
A. refine.bio [5]	web service	non-programmer	bulk RNA-seq, microarray	N, rule-based harmonisation	N	N	N, AWS Batch	n.r.	n.r.	none	N
A. ARCHS4 [4]	web and Python package	both	bulk RNA-seq, human and mouse	N	N	N	N, vendor cloud	n.r.	n.r.	none	N

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
A. recount3 [3]	R package and web	programmer	bulk RNA-seq, human and mouse	N	N	N, Monorail can run locally	P	n.r.	n.r.	none	P
A. MetaSRA [33]	pipeline and web database	programmer	SRA sample metadata	N, ontology mapping and classifiers	N	n/a	local	n/a	n/a	none	Y
A. Celline [34]	Python CLI and web UI	both	single-cell	Y, LLM metadata standardisation	N, manual edits	N	local or PBS cluster	n.r.	n.r.	OpenAI	Y
B. pysradb [10]	CLI and Python	programmer	metadata and download, any assay	N	N	N	local	n.r.	n.r.	none	Y
B. ffq [9]	CLI	programmer	metadata and links, any assay	N	N	N	n/a, no download	n/a	P, md5 when the source provides it	none	Y
B. nf-core/fetchngs [13]	Nextflow pipeline	programmer	any assay	N	N	N, fixed	wherever Nextflow runs	P, Nextflow retries, no hand-over	Y, md5 on some download paths	none	Y
B. Kingfisher [35]	CLI	programmer	any assay	N	N	N	local	P, falls through download methods	n.r.	none	Y
C. ASReview [17]	local web app	non-programmer	literature	N, active learning without an LLM	Y, human labels	N	local	n/a	n/a	none	Y
C. Rayyan [18]	hosted web and mobile	non-programmer	literature	Y, AI reviewer with notes	Y	N	vendor cloud	n/a	n/a	vendor-managed	N

Tool	Form	Target user	Data scope	LLM discovery or curation	Human review with evidence	User-defined pipeline	Runs on user's own server	Bounded retry, then hand-over	Checksummed outputs	LLM backend	Local, no server to deploy
C. Covidence [36]	hosted web	non-programmer	literature	P, classifier screening, LLM extraction	Y, supporting quote from PDF	N	vendor cloud	n/a	n/a	single vendor	N
D. Galaxy [16]	web platform on a server	non-programmer	any assay	N	n/a	Y, graphical editor	admin-configured clusters	P, resub-mission rules	n.r.	none	P, needs a running server
D. Seqera Platform [37]	hosted or self-hosted web	both	any assay	P, pipeline-development assistant	n/a	Y, any Nextflow pipeline	HPC over SSH or agent, cloud	Y, monitoring and Nextflow retries	n.r.	vendor-managed	N
D. Snakemake [14]	CLI and Python DSL	programmer	any assay	N	n/a	Y	local, cluster, cloud	P, retries, no hand-over	n.r.	none	Y

Table 6 reads the same comparison the other way round: for the closest work, what it already provides and what this paper adds. Where a preceding system already provides a mechanism, this paper does not claim it.

Table 6: **The closest work placed against this paper's question.** Each cell reports what the source's documentation or paper explicitly describes; "not reported" in Table 5 is a literature boundary, not a claim of impossibility.

Closest work	What it already provides	What this paper adds
scBaseCount and SRAgent [6]	Agentic discovery and metadata extraction over the SRA, uniform reprocessing, the largest single-cell repository, and a stated blueprint	The blueprint as a general local tool: free-text scope, the user's own pipeline and server, human gates before processing, verified evidence, agents invoked rather than resident
Uniformly reprocessed resources: recount3, ARCHS4, refine.bio [3,4,5]	Ready-made, centrally processed expression compendia for bulk assays	A way for a laboratory to make a compendium of its own scope and assay with its own tools, and to keep extending it
Dataset fetchers: pysradb, ffq, nf-core/fetchngs, Kingfisher [13,9,10,35]	Metadata retrieval and download for accessions that have already been chosen	The choosing: search, agent-suggested review with evidence, recorded decisions; and what happens after the download
Screening tools: ASReview, Rayyan, Covidence [17,18,36]	Human-in-the-loop screening of literature with learned or LLM suggestions	The same gate applied to datasets, followed by processing and archival in the same application
Workflow platforms: Galaxy, Seqera Platform, Snakemake [14,16,37]	Execution, scheduling and retry of pipelines on diverse compute	The layer above the engine: which datasets to run, why they were included, and whether each result arrived intact; an engine can be called from a step
Agentic metadata curators [30,31,32]	LLM curation of GEO metadata at repository scale, in one case with triage for human review	Curation tied to the decision, the processing and the archive, in a desktop tool rather than a pipeline
Static data publishing: Datasette, GitHub Pages, Cloudflare Pages [23,21,22]	Serving a table as a browsable site with an API; free static hosting with version control and a deployment interface	A domain skill that guides the agent in inspecting the supplied data and authoring a site's structure, visualisations and interactions; a standalone page with embedded data and agent-written code, with templates optional; provider-mediated publication is separate. Dynamic sites for large datasets remain future work

3 Discussion

scBaseCount demonstrated that the loop of discovering, curating, reprocessing and archiving public data can be run by agents, and it offered the approach as a blueprint [6]. DBA takes the blueprint at its word and asks what it looks like as a tool rather than a project. The answer is smaller in every dimension that matters to a laboratory: one binary instead of a cloud deployment, free-text criteria instead of an assay-specific classifier, shell steps instead of a bespoke Nextflow pipeline, the lab's own server instead of a batch service, and a recorded decision at three gates instead of an autonomous supervisor. The two systems are not competitors. scBaseCount is a repository; DBA is a way of making repositories, including ones scBaseCount will never contain, such as GPSAdb's human knockdown and knockout compendium, a plant

tissue atlas, a proteomics cohort or a set of imaging studies whose processing is a command the user already knows how to type.

Four design decisions deserve comment. First, verifying evidence by substring is a cheap guard that changes the character of the review step: a suggestion that cannot point to the words it relied on is demoted before anyone reads it, and a weak or small model fails safely. It also sets a floor on what the reviewer can be asked to do, since only the title and abstract are available; scBaseCount notes the same limit when study-level labels are propagated to samples [6], and full-text or per-sample metadata is the obvious next source. Second, sparse agents and the separation of orchestration from computation make the system restartable and cheap. Nothing in the desktop process holds state that is not already on disk, and nothing on the server depends on the desktop being open; the cost of the system between decisions is one SSH call per job per interval. Third, agents act, but only through the application's own command layer, and the restriction is enforced in code rather than in prompts: the tool list is fixed per agent, every command passes the same validation, membership checks and rollback as a form, decisions store their author, and no tool runs a shell or reaches the executor. The first release kept agents read-only; letting them act is what allows the assistant to carry a goal through the workflow, and it moved the safeguard from the tool list to the command layer. Indirect prompt injection through fetched abstracts and logs is a known risk for tool-using models [38]; an injected instruction can now at most issue a command the user could have issued, in a project the agent is scoped to, with the agent recorded as its author, and the prompts require an explicit instruction before any change is made. What is not verified is a decision recorded from the conversation: the evidence check applies to batch suggestions, and a chat-recorded decision is a model judgment with a reason. Fourth, maintenance is designed in rather than hoped for: locked versions, per-attempt directories, idempotent submission, manifests with the review decision and sample groups beside the output list, and merge-on-search mean that a repository can be extended next year by the same project without re-deciding what was decided.

Website construction follows the same principle: a domain skill is readable, editable knowledge that guides the agent's decisions, while the resulting interface is written for the actual data and user goal. The optional table/card vocabulary does not bound Agent-authored interfaces. The PBMC5k case demonstrates this separation for processed single-cell results. It also creates a validation obligation: generating more flexible code can introduce interaction errors or misleading views, which input checks and a sandbox alone cannot detect.

The intended design includes both static and dynamic sites. Browser-resident static output is the first demonstrated implementation; it avoids an application backend for suitable datasets. Large matrices, linked tables, concurrent users or ongoing updates may instead need a database, query API and server-side computation. Selecting and operating that architecture is a further Agent capability to develop and evaluate, not a capability established by the PBMC5k example. MCP provides a route to configured hosting services, but a trace of an external call is not evidence that a particular site has been deployed correctly or can be maintained.

Using shell steps as the pipeline language is a deliberate trade. It is the lowest barrier, since every bioinformatician already has the commands, and the most general, since any tool, container or workflow engine can be invoked from a step; a Nextflow or Snakemake run is one line. What it gives up is what those engines provide inside a job: dependency graphs, caching, per-process resources and cloud back ends [8,14,15]. DBA is therefore best understood as the layer above the workflow engine, concerned with which datasets to run and whether they ran, not with how a run is parallelised.

3.1 When the claim does not hold

Several observations would defeat the contribution as stated:

- If a reviewer without the evidence check includes and excludes as accurately as one with it on a reference set such as GPSAdb's curated membership or SRAgent's labels [6,11], the verified-evidence gate is overhead rather than a safeguard.
- If the pipelines laboratories actually run cannot be written as shell steps over the six contract variables without placing a workflow engine inside every step, the generality claim reduces to a job submitter for one engine.
- If sparse invocation and the tick model make unattended processing of a project slower or more failure-prone than a resident supervisor by more than the model cost they save, the sparse-agent design is a false economy.
- If maintaining a repository across several pipeline versions still needs bookkeeping outside the version snapshots and manifests, the maintenance claim fails.
- If a researcher who does not program cannot complete a project without help, the usability claim fails; no observation of this has yet been made.
- If domain-guided agents cannot generate correct, usable interfaces from unfamiliar result schemas without substantial programmer repair, the autonomous-authoring claim is limited. If larger datasets require backend design or maintenance that the agent cannot reliably perform, dynamic website construction remains an unmet part of the design.

These criteria also name the work needed: build a repository end to end on a real server, measure review agreement against a reference, and observe researchers using the tool on their own questions.

3.2 Costs and risks

The design carries concrete costs. Review from titles and abstracts alone will miss datasets whose relevance is only visible in full text or sample attributes, and a wrong exclude is silent, since the evidence check guards what is included rather than what is dropped. Shell steps give no dependency graph or caching, so a laboratory that needs them must bring an engine. Three human gates cost time in proportion to the number of candidates, which is the point, but it bounds the scale at which DBA is pleasant to use. The evidence check and the fixed tool lists limit what an injected instruction can do, but a persuasive message can still mislead the person reading it. State in one JSON file is simple to back up and inspect but will not scale to a repository of millions of candidates. Finally, a tool that runs on the user's own server inherits that server's reliability, and DBA's recovery model assumes the server comes back; it does not provide an alternative. The static kind adds two dependencies outside the application: the hosting provider's limits, which on 6 November 2026 were a 1 GB site, a soft 100 GB of monthly bandwidth and 10 builds per hour for GitHub Pages [21], and 20,000 files of at most 25 MiB each and 500 builds per month on Cloudflare's free plan [22]; and the provider's MCP server, whose tools can change underneath DBA. Deployment depends on the configured provider tools and permissions; a recorded tool call does not establish that the resulting site works or that a build-specific deployment review has been completed.

3.3 Limitations

The current website implementation imports one table of at most 2 MiB, 10,000 rows and 100 columns and exports one self-contained HTML document. These are implementation limits, not measured browser scalability thresholds. Agent-authored code supports more than two fixed layouts, but all data still reside in the page; raw H5AD, RDS and FASTQ processing is outside this authoring path. Generated JavaScript and scientific interpretations may be wrong even when structural validation passes. The PBMC5k run validates one live-model construction and

a defined set of browser interactions, not reliability across datasets or models; direct offline file opening and public deployment remain unverified.

Large-scale dynamic websites are a current limitation within DBA's intended scope. Large expression matrices, cross-table queries or data that exceed browser memory and rendering capacity may require indexed storage, backend APIs, on-demand or chunked transfer and server-side computation. Multi-user permissions and continuous updates introduce additional deployment and operational requirements. DBA does not yet provide an end-to-end Agent workflow for generating, validating, deploying and maintaining this stack, and no dynamic-site performance or concurrent-access benchmark has been conducted. Moving from a collecting project's archive to website input also remains a manual export/import step. The executor supports Linux-style hosts over SSH only; Slurm or other schedulers, containers, memory and disk quotas, and cancellation or deletion of remote jobs are not provided, and the CPU quota governs admission rather than the threads a tool actually uses. State is one JSON file, suitable for thousands of candidates and hundreds of jobs per workspace, not for millions. Three sources are built in and others arrive by JSON import. Review uses titles and abstracts only, and a decision recorded from the conversation is not checked against evidence. The interface switches between Chinese and English, but the agent instructions and the review prompt are in Chinese in the shipped presets, though both are editable text. No measurement of review accuracy against a gold standard, of time saved, or of usability has been made. One dataset has been carried end to end on a loopback host; no multi-dataset repository has been completed and no remote server has been used in those collection cases. The separate PBMC5k case used a live model for website authoring, not for an end-to-end collection workflow. These examples do not establish that the whole system works for a research group. The binaries are unsigned. Agents can act, which means a misled or misinstructed agent can record a wrong decision; the application keeps the author and reason of every decision but does not undo one.

3.4 Future work

A central direction is to extend data-driven Agent authoring to large-scale dynamic sites. The agent should use data volume, schema, query patterns, update frequency and access requirements to choose between browser-resident output and a database-backed architecture. Domain skills would guide interpretation and useful interactions, while additional tools would support schema/index design, API construction, frontend data access, deployment and ongoing monitoring. Evaluation must test correctness, query latency, browser and server memory, concurrent access, failure recovery and operating cost on progressively larger datasets; increasing an import limit alone would not establish scalability.

Other priorities are automated browser checks of generated interfaces, repeatable domain-specific authoring evaluations, deployment manifests and build-specific approval records, and a direct bridge from archived analysis results to website inputs. Collection priorities include Slurm and container-aware execution, further metadata sources, full-text evidence retrieval, and evaluation against GPSAdb's manually curated membership followed by an automated update of that resource. These remain development and evaluation goals rather than implemented features. Two further directions follow from the design. A project definition, which is criteria, search expressions, pipeline versions and output contract, is a small text object that could be shared as a recipe so that another group can rebuild the same repository against the same sources, which would make the repositories reproducible as well as the analyses. And because every decision carries a reason and every job a manifest, a workspace is already an audit trail; exporting it in a FAIR-compatible form would let a DBA repository be published as a resource in its own right [39].

4 Methods

4.1 Implementation

DBA is written in Go 1.26 and has four direct dependencies: Wails v3.1.0 [40], the official MCP Go SDK [28], a small library that reads and writes the operating system's credential store, and a YAML parser for skill files. The front end is plain HTML, CSS and JavaScript embedded in the binary with Go's embed facility and served by the same handler that serves the JSON API, so the desktop window and a development browser session see identical behaviour. The handler binds to the loopback interface, rejects requests whose Origin host differs from the request host, sets no-store and nosniff headers, and limits POST bodies to 4 MiB. Builds are driven by one shell script and one PowerShell script that accept windows, darwin, linux or all as targets; the Windows installer is produced by NSIS 3 and the macOS image by hdiutil.

```
scripts/build.sh all
```

```
go test -race ./... && go vet ./... \
  && node --check frontend/workbench.js frontend/conversations.js \
  && node --test scripts/test-conversations.js
```

4.2 Installation and first use

Release builds target all three platforms from one machine, Windows as a pure-Go target and macOS or Linux through Wails' pure-Go backend when cross-compiling, and the installers need no administrator rights (Table 7). The workspace is a folder in the user's configuration directory, selectable with a flag, and one desktop instance runs per workspace.

Table 7: **Distribution.** Sizes are the unsigned builds of 5 November 2026, before the desktop icon and subsequent website-authoring changes. These historical measurements do not describe a current release build.

Platform	Runtime dependency	Artefact	Size
Windows 10 or 11	WebView2 Runtime (bundled with Windows 11)	DBA.exe; DBA-Setup.exe (per-user NSIS installer with uninstaller)	11.7 MB; 3.7 MB
macOS 11 or later	none (system WebKit)	DBA.app; DBA.dmg	9.7 MB; 4.6 MB
Linux	GTK 4 and WebKitGTK 6.0, or GTK 3 and WebKit2GTK 4.1 with a build tag	DBA	11.1 MB

First use takes three settings. A model profile is a protocol, a base URL, a model identifier and a key entered on the preferences page and kept in the system credential store; a local Ollama or vLLM endpoint needs no key. A server is a host, port, user, optional key file, an absolute dedicated work directory, a CPU quota and a job limit; the connection check creates the directory, confirms it is writable, and reports the kernel, the CPU count, free disk space and the presence of nohup, sha256sum and realpath. Agents are then assigned a profile each, or inherit the first.

4.3 Workspace

A workspace is a directory holding `workbench.json` and `projects/<project>/results/<job>/`. The JSON document contains projects, servers, jobs, artefacts, conversations, skills, events and

settings, including the MCP server records, under a schema version. Writes serialise the whole document to a temporary file in the same directory, fsync it and rename it over the previous file. The event list is trimmed to the latest 2,000 entries; conversations are not trimmed. Each completed job writes `run_manifest.json` with the schema version, job and dataset identifiers, source accession and URL, organism, review decision and note, sample groups, the full pipeline snapshot, server identifier, attempt number, output file list with SHA-256 and sizes, and a completion timestamp. A project manifest export bundles the project, its jobs and its artefacts with server key paths removed.

4.4 Discovery

Europe PMC is queried through its REST search endpoint with resultType core, 100 records per page and cursor-mark paging until the requested limit, at most 1,000, is reached. PubMed and GEO are queried through Entrez: esearch with JSON output and a retmax equal to the limit, then efetch in XML for PubMed records or esummary in JSON for GEO DataSets, in batches of 100 identifiers separated by 400 ms. Requests carry an identifying tool name and user agent, time out after 90 s, and read at most 20 MiB. Titles and abstracts are stripped of HTML tags. Candidates merge into the project by three keys: whitespace-normalised lower-case title, DOI without prefix, and accession; a record matching any existing key is dropped and the search record stores hits, fetched, added and timestamp. Imports accept 1 to 1,000 records per request and are merged by the same keys.

4.5 Review

The reviewer's batch mode sends, per candidate, the project name, criteria, organism and assay and the candidate record, with a system prompt that declares the candidate text untrusted and requires a JSON object with suggestion (include, exclude or uncertain), confidence (high, medium or low), reason and evidence. A reply that is not valid JSON or has an invalid suggestion is rejected and nothing is stored. A reply whose evidence is empty or is not a substring of the title plus abstract is stored as uncertain with low confidence, no evidence and a note asking for manual inspection. At most 20 pending candidates without a suggestion are processed per request. A human decision requires a non-empty reason; accepted candidates may carry an accession, a download URL that must be HTTP(S) without embedded credentials, and sample groups given as one sample and group per line with at least two samples and two groups.

4.6 Agent loop

A chat message of at most 8,000 bytes is enqueued in a conversation of the project, or of the workspace when no project is selected, with a client-generated identifier that makes a repeated submission idempotent; the service runs the messages of a scope in order, one at a time, with at most twenty waiting, and sends the last sixteen messages of completed turns in that conversation. The system prompt is the agent's instruction, the guardrail block, and the project name, identifier and current time. On each step the model receives the conversation and the agent's tool definitions; every returned call is validated against its schema (required fields, types, enumerations, numeric bounds, unknown fields), executed in order against the latest committed state, and its JSON result (or a JSON error) is appended as a tool turn, while a trace entry records tool name, arguments clipped to 8,000 bytes, result clipped to 16,000 bytes, an error flag and the elapsed milliseconds. Results above 16,000 bytes are truncated with a flag; list tools accept an offset and a limit of at most 50. Command tools call the service's dispatcher with the same command objects as the forms; a partial update preserves omitted fields, a supplied array replaces the whole list, and an invalid command rolls back. The loop runs for at most the

agent's step budget; if the budget is spent, a final call is made with tool use disabled and a user turn stating that the budget is exhausted. Tool names outside the agent's list are refused with an error returned to the model. A mention of a specialist in the message, or a delegate call by the project assistant naming a specialist and a task, runs that specialist's own loop with its own model, instructions, tools and budget over the shared history and a fresh snapshot; its reply, trace and failure status are stored as a task card in the same conversation and returned to the assistant as a tool result. Specialists cannot delegate, a turn allows at most eight delegations, and a message sent while another is running in the same project waits in the queue, where it can be edited or withdrawn; a stop or a failure pauses the queue, and a restart marks the running request as interrupted with its completed tool calls preserved and nothing replayed.

4.7 Model adapters

All three adapters translate the same turn list. The OpenAI-compatible adapter posts to `base/chat/completions` with a bearer token and function tools, and sets tool_choice to none on the final call. The Anthropic adapter posts to `base/messages` with the x-api-key header, API version 2023-06-01 and max_tokens 8192, groups all tool results of one assistant turn into one user message, keeps the tool definitions in every request because the history contains tool blocks, and sets tool_choice type none on the final call. The Gemini adapter posts to `base/models/{model}:generateContent` with the x-goog-api-key header, sends function declarations, matches function responses by name, and sets the function-calling mode to NONE on the final call. Responses are capped at 4 MiB. Proxies follow HTTP_PROXY and HTTPS_PROXY. Keys are read from the operating system's credential store at call time; profiles from the first release may still name an environment variable.

4.8 Scheduling and execution

A ticker fires every 2 s; a tick runs when the configured interval (default 60 s, range 10 s to 86,400 s) has elapsed and the scheduler is not paused. The tick first promotes retry-wait jobs whose retry time has passed back to queued, then launches observation of starting, running and syncing jobs, then admits queued jobs while the server's CPU quota, the server's job limit and the project's concurrency are respected, with at most eight operations in flight. Each admitted job increments its attempt counter and receives the remote directory `<workdir>/<project>/<job>/attempt-N`. Start writes `run.sh` into that directory inside a lock directory and behind a submitted marker, then launches it with nohup and records the process id; the script exports DBA_ACCESSION, DBA_SOURCE_URL, DBA_INPUT, DBA_OUTPUT, DBA_SAMPLE_GROUPS and DBA_CPUS, creates the input and output directories, runs the steps in a subshell under `set -e` from the output directory, and writes the exit code through a temporary file and rename. Poll reports EXIT with the code if the exit file exists, RUNNING if the recorded process answers `kill -0`, otherwise UNKNOWN, and returns the last 64 KiB of the log. Transient failures are recognised case-insensitively by the pattern `curl: \((6|7|18|28|35|52|56)\)|connection (reset|timed out|refused)|temporary failure|network is unreachable` and retried after 10×2^n seconds capped at 300 s while the attempt count is within the configured maximum (default 3, range 0 to 10). An SSH-level error during any operation increments a failure counter, keeps the job's state and schedules the next attempt with the same back-off; when the counter exceeds the maximum the job moves to needs-help with its current state as resume point. Operations time out after 45 s, synchronisation after 30 min and a server probe after 30 s. SSH is invoked as the system client with BatchMode, StrictHostKeyChecking yes, a 12 s connection timeout and keep-alives, and with the configured key file if any.

4.9 Synchronisation

For each declared output, a remote script resolves the real path of the output directory and of the file and exits if the file is outside the directory, then prints the SHA-256 and byte count. If a local file with the same digest already exists it is reused. Otherwise the file is streamed from the byte offset of any partial local copy with `tail -c`, written to a partial file, fsynced, hashed locally and renamed into place only if its digest matches; on mismatch the partial file is truncated and the job waits for the next tick. Files are limited to 100 GiB. Output names must be relative, contain only letters, digits, dot, underscore, dash and slash, and contain no empty, dot or dot-dot components.

4.10 Skills

A SKILL.md is UTF-8 text of at most 12,000 bytes: a front matter delimited by lines of three dashes with name (1 to 64 lower-case letters, digits and hyphens), description (at most 1,024 bytes) and an optional metadata.dba block with label (at most 120 bytes), layout (table or cards), accent (a six-digit hexadecimal colour, default #237c63), required-columns (at most 100 distinct names, checked for presence only) and title-column (required for cards; when given, it must exist in the input); the body after the front matter must not be empty. A file without the metadata block is an ordinary instruction skill and does not appear in the template selector. An import is refused for a malformed front matter, a duplicate name or a fifty-first skill and can guide custom-site authoring. The three shipped skills are installed once into a workspace that has no skills field; an emptied list stays empty. Enabled skills appear in every agent's prompt as name and description; the read tool returns the full file and the parsed template with a result limit of 80,000 bytes, so a skill is never truncated; the list and read tools are added to every specialist's tool list.

4.11 Static sites and MCP

Both website routes are restricted to static projects. Input parsing accepts UTF-8 CSV, TSV or an array of JSON objects, up to 2 MiB, 10,000 rows and 100 columns; ragged rows and empty or duplicate column names are rejected. Fixed-template builds validate required columns and render through Go's HTML templates. For custom authoring, a separate import command saves the input; `inspect_site_data` returns column-level summaries and a bounded sample (five rows by default, at most 100), with category counts for columns having at most 30 distinct values.

`build_custom_site` accepts an enabled skill identifier, title, description and a complete HTML document up to 512 KiB. Titles and descriptions are limited to 200 and 4,000 bytes. The service parses HTML, rejects external resource attributes and unsupported active elements, inserts a CSP before scripts, and injects the complete input through `window.DBA_DATA` with JSON encoding that escapes HTML-sensitive characters. Inline JavaScript/CSS and data images are permitted; external resource loading and network APIs are constrained by the policy. Model requests allow five minutes while retaining context cancellation; discovery retains its 90-second timeout.

The site snapshot stores data and generated code separately from later input imports. Validation and storage are transactional, and failed rebuilds preserve the previous site. Preview uses both an HTTP sandbox policy and an iframe without same-origin permission. The same renderer serves the download with an attachment header; the exported HTML retains the embedded CSP. These mechanisms constrain execution and preserve data but do not prove functional or scientific correctness.

An MCP server record holds a name, a transport (Streamable HTTP or STDIO), a URL or a command with arguments and named environment variables, an enabled flag, an optional bearer-token variable name or credential reference, a startup timeout (default 20 s, at most 120 s), a tool timeout (default 60 s, at most 600 s) and optional allow and deny lists, of which the deny list wins. An HTTP URL may carry no query, fragment or credentials and must be HTTPS off loopback; the HTTP client sends the bearer token, refuses redirects and caps bodies at 8 MiB and server-sent events at 4 MiB. A STDIO server is started in the workspace directory with an environment of basic system, locale and proxy variables plus the named ones. Tool listing follows cursors for at most 100 pages and 1,000 allowed tools. The tool-listing tool returns at most 20 tools per call with the server's instructions clipped to 4,000 characters; the call tool requires a listed tool and an object of arguments; results are redacted of secrets, marked failed when the server sets its error flag, and truncated above 16,000 bytes. A connection is reused within one agent turn and closed after it; a connection test uses its own session. A token saved in preferences goes to the credential store under a reference that the state API never returns; changing a server's URL while a token is saved requires the token to be re-entered or removed.

4.12 Tests

The Go test suite is run with the race detector; a separate Node test script covers the conversation front end. Scheduling tests substitute a controlled executor that scripts probe, start, poll and sync outcomes; model tests substitute an HTTP test server that returns fixed responses for each protocol, including scripted tool calls and delegations; discovery tests substitute HTTP responses for the three sources; credential tests use the keyring library's in-memory mock. MCP tests run Streamable HTTP, server-sent-event and STDIO servers inside the test process, including through the full agent loop under all three model protocols; skill and site tests build sites from CSV, TSV and JSON and fetch the preview and the download over the API.

5 Data and code availability

DBA source code will be publicly released after this preprint is posted. The accompanying supplementary materials contain the recorded case-study inputs, scripts, generated outputs and verification records. The single-dataset collection case includes the candidate list, processing commands, sample assignments and run evidence with checksums under docs/manuscript/gpsadb-case/. No new biological data were generated. The count matrix archived in that case is a standardised copy of the submitter-provided table of GEO series GSE299540.

The PBMC5k input table, source checksums and Agent-authored page code are under examples/pbmc5k/; screenshots, tool traces, exported HTML and verification limits are under docs/walkthroughs/pbmc5k/. Its 10x Genomics source data are reused under CC BY 4.0. The supplementary source package contains the manuscript and the figure files needed to compile it.

6 Author contributions

Guihui Qin developed and tested the software. Zhou-Geng Xu provided the software development concept and design direction. Both authors approved submission of the manuscript.

7 Acknowledgements

This work received no specific funding. The authors thank the Europe PMC and NCBI teams for public programmatic access to their services.

8 Declaration of interests

The authors declare no competing interests.

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